Whole-Teflon microfluidic chips

Kangning Ren, Wen Dai, Jianhua Zhou, Jing Su, and Hongkai Wu[1]

Department of Chemistry, Hong Kong University of Science and Technology, Clear Water Bay, Kowloon, Hong Kong

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Although microfluidics has shown exciting potential, its broad applications are significantly limited by drawbacks of the materials used to make them. In this work, we present a convenient strategy for fabricating whole-Teflon microfluidic chips with integrated valves that show outstanding inertness to various chemicals and extreme resistance against all solvents. Compared with other microfluidic materials [e.g., poly(dimethylsiloxane) (PDMS)] the whole-Teflon chip has a few more advantages, such as no absorption of small molecules, little adsorption of biomolecules onto channel walls, and no leaching of residue molecules from the material bulk into the solution in the channel. Various biological cells have been cultured in the whole-Teflon channel. Adherent cells can attach to the channel bottom, spread, and proliferate well in the channels (with similar proliferation rate to the cells in PDMS channels with the same dimensions). The moderately good gas permeability of the Teflon materials makes it suitable to culture cells inside the microchannels for a long time.

microchips | on-chip cell culture | solvent resistive chip

This report describes a convenient method for the fabrication of whole-Teflon microfluidic chips; we also integrate microvalves into the chips and demonstrate a few key applications of the whole-Teflon microfluidic chips with various organic solvents and biomolecules, and for culturing biological cells.

Microfluidics (1) emerged during the early 1990s with channel networks in silicon or glass. Microprocessing of these materials is labor-intensive and time-consuming, it requires sophisticated equipment in a clean room, and often involves hazardous chemicals. The subsequent use of poly(dimethylsiloxane) (PDMS) greatly simplified the fabrication of microchips (2) and led to the rapid development of the field. PDMS has other attractive properties, such as being elastic (easy to make efficient microvalves), permeable to gases, and compatible with culturing biological cells. Despite these advantages, applications of PDMS chips are severely limited by a few drawbacks that are inherent to this material: (*i*) strong adsorption of molecules, particularly large biomolecules, onto its surface (3); (*ii*) absorption of nonpolar and weakly polar molecules into PDMS bulk; (*iii*) leaching of small molecules from PDMS bulk into solutions; and (*iv*) incompatibility with organic solvents. Therefore, special attention must be paid when quantitative analysis is needed (materials lost on channel walls and into the PDMS bulk) or organic solvents are involved. Because many drug molecules are small with relatively low polarity and cells can be very sensitive to their environment, data interpretation requires particular caution when cell-based drug screening and cell culturing are performed on PDMS microchips. Various techniques (4–8) have been proposed to modify PDMS bulk and surface properties, but they normally complicate the fabrication and still cannot effectively solve the problems. Most other plastics (9–11) have similar problems as PDMS. Alternatively, Quake and coworkers fluorinated silicone polymer chains as a chemically inert chip material (12). However, because the polymer is only partially fluorinated, many solvents can still swell the material. Moreover, it is neither commercially available nor easy to synthesize. A recently reported partially fluorinated plastic (Dyneon THV, a co-polymer of tetrafluoroethylene, hexafluoropropylene and vinylidene fluoride) (13) is not resistant to organic solvents either.

To overcome all these problems, Teflon plastics seem to be the perfect solution. Teflon is a DuPont registered trademark for several perfluorinated polymers; they are broadly used in various processes that involve quantitations of trace amounts of chemicals and require anticorrosion and superclean environments. They are well-known for their superior inertness to almost all chemicals and all solvents; they also show excellent resistance to molecular adsorption and molecule leaching from the polymer bulk to solutions (14, 15). Among them, the two semicrystallized polymers, perfluoroalkoxy (PFA) and fluorinated ethylene-propylene (FEP), are suitable for microfluidic chips because of their optical transparency and proper mechanical strength (16) (which enables the fabrication of diaphragm valves). There are two major challenges. First, no current method can conveniently generate micropatterns in PFA and FEP (17). Reported micropattern techniques such as X-ray lithography, electroplating, and molding (18), synchrotron radiation (19), and ion beam writing (20) are either too expensive or require serial processing. Because PFA and FEP are melt-processible, microfeatures can be transferred into them with thermo molding using micropatterned masters in silicon, glass, and metals (21–23). However, air is often trapped during the molding and very limited types of microfeatures can be formed. The high melting temperatures of PFA and FEP (both over 260 °C) preclude the use of most plastics as the master. For example, Hira, et al. tried to mold polytetrafluoethylene with a polycarbonate (PC) (Tg ~ 150 °C) master at low temperatures (<150 °C); the fidelity of the molding is quite low (e.g., sharp edges of microfeatures are severely rounded) and the PC master is prone to damage each time under high molding pressure (24). Second, PFA and FEP are soft (especially around their melting temperature) with sharp transition temperature ranges; the pressure during thermobonding can easily crush the channels.

Herein we describe a convenient method for the fabrication of whole-Teflon microchips. First, we develop a high-temperature (up to 350 °C) thermo molding technique using specially prepared PDMS masters that can pattern Teflon plates with extremely high resolution and fidelity. The molding of a Teflon microchannel takes approximately 5 min on a common hot compressor without using clean room or involving expensive equipment and harsh chemicals. Second, we introduce a convenient method to seal whole-Teflon microchannels with extremely high yield. Third, we demonstrate whole-Teflon chips that integrate monolithic pneumatic valves and pumps. Finally, we evaluate the performance of the Teflon chips by examining their compatibility with various solvents, biomolecules, and cell cultures. With our method, various complex microstructures can be generated in the Teflon chips; the fabrication process is convenient and amenable for mass production. In addition, the Teflon plates can be recycled for multiple uses without risk of contamination.

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[1]To whom correspondence should be addressed. E-mail: chhkwu@ust.hk.

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Results and Discussion

To find a suitable material for the master of thermo molding Teflon materials, we tested many polymers and found that they degrade and/or deform severely when temperature is above 260 °C. PDMS is widely used as the master material for pattern transfer in soft lithography (2); however, the highest operating temperature of PDMS for thermo molding is reported to be approximately 150 °C [gas bubbles start to form in the molds above this temperature (25) probably because of the release of small molecules from the PDMS bulk]. Our initial experiments confirmed the generation of gas bubbles at high temperatures during thermo molding and failed to transfer micropatterns to Teflon. Normal PDMS is produced by mixing the two precursor components of PDMS at the recommended weight ratio (A : B = 10 : 1). We found that by changing this ratio and with proper treatment we can greatly raise its operation temperature. Fig. 1 *A* and *B* illustrates the rapid prototyping technique to mold microchannels into Teflon with this PDMS. To obtain the highest operation temperature, we adjust the mixing ratio of A and B to 5 : 1 and treat the PDMS in two consecutive steps: Cure the PDMS at 70 °C for 30 min and peel the PDMS master off from the photoresist structures and bake it at 250 °C for 1 h. This treated PDMS master can be used for thermo molding at temperatures up to 350 °C without failure. Fig. 1 *C–F* shows some representative PFA channels formed (both square and rounded sections). At the high molding temperature, the gas permeability of the PDMS master ensures fully filling of the microfeatures during the molding; its elasticity facilitates the release of the mold after the molding. As a result features as small as approximately 100 nm can be transferred into the Teflon materials with high fidelity (Fig. S1). Because it is very easy to generate various microstructures in PDMS, different microstructures can be formed in these Teflon materials with this single-step molding technique. Therefore, in contrast to glass or silicon microchips, we are free to design microchannels and microstructures on Teflon chips with wanted shapes and depths.

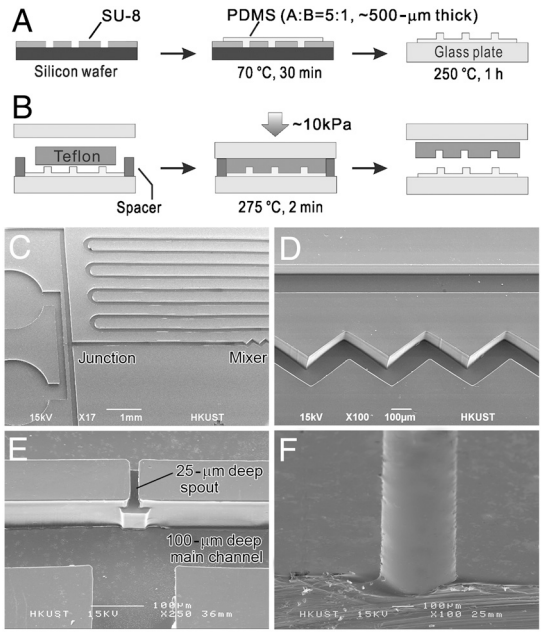


Fig. 1. Molding of Teflon microchannels using PDMS masters. (*A*, *B*) Schematics of the preparation of thermally stable PDMS masters and the molding of Teflon channels, respectively. (*C–E*) SEM images of microfabricated PFA channels (*C*) for droplet generation that includes a micromixer (*D*) and a 3-D spout (*E*). (*F*) Microchannel with a rounded profile molded from a reflowed positive photoresist (AZ4903) structure.

Fig. 2 shows the convenient sealing of Teflon microchannels. The Teflon channels are sealed with normal thermobonding technique, but both temperature and pressure needs cautious controls. Because the transition ranges of temperature for PFA and FEP changing from solid to liquid are narrow, the bonding temperature needs to be carefully tuned. Another key factor for successful sealing of Teflon channels is the way of applying pressure. Without pressure, the two Teflon plates do not bond to each other at any temperature. However, pressure constantly applied through weight or binder clips can easily destroy the channel. At the bonding temperature, the Teflon materials are very soft so that even a low pressure can severely deform the channels. We found that screw clamps of stainless steel give the best control of pressure for the bonding. Because PFA (and FEP) has a larger linear expansion factor (310 ppm/°C) than the clamps (11.8 ppm/°C), pressure is spontaneously generated during heating from room temperature when the screws are fixed. The two PFA pieces are pressed hard and bonded to each other at the bonding temperature. The bonding formation releases the pressure, which greatly reduced the possible deformation of the microchannels. Under the optimized conditions, the yield of this bonding process was approximately 100% for all 20 microchannels in the size of 20–200 μm that are tested. Microchannels with low aspect ratios (~1 : 5) are easy to collapse during the bonding. This process also enables us to bond multilayer chips in one step (Fig. 3*A*), which is important for fabricating integrated diaphragm valves.

As expected, the Teflon chips show excellent resistance to organic solvents, which swell or dissolve other polymers used for microchips (3). Fig. 3*A* shows a microchip filled with dyes in organic solvents in a two-layer microchannel device. Fig. 3*B* shows laminar flows of hexane, chloroform, and toluene in a PFA chip. No swelling of the channel can be observed after 24 h or longer. We also studied the resistance of our Teflon chips against adsorption of biomolecules. We fill the Teflon channel with a 100 μg/mL aqueous solution of GFP and incubate for 10 min (Fig. 3*C*). After rinsing the channel with buffer, no fluorescent signal against background can be seen on the wall of the channel under a fluorescence microscope. In contrast, under the same conditions, a significant amount of the protein is observed on the walls of a PDMS channel and a polystyrene (PS) channel

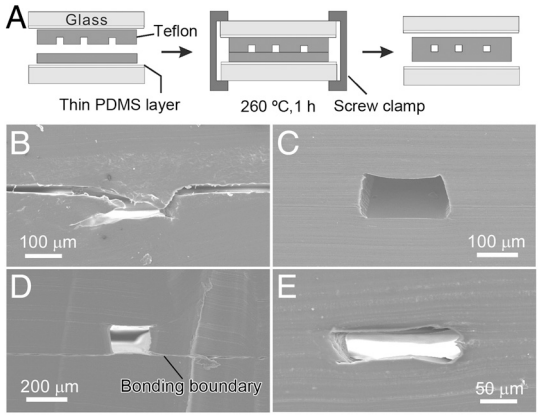


Fig. 2. Schematics of the bonding of Teflon chips (*A*) and SEM images of cross-sections of channels sealed with constant pressure (*B*) or screw clamps (*C–E*). (*A*) The bonding assembly is fixed and tightened by screw clamps before being heated in an oven for bonding. (*B*) Channel bonded at lower temperature (245 °C) with constant pressure of approximately 100 kPa. At higher bonding temperatures (same pressure), the two Teflon plates merge completely. (*C*) Channel bonded at the optimized temperature (260 °C) with screw clamps. (*D*) Channel bonded at lower temperature (250 °C) with screw clamps. Bonding boundary can be clearly seen and the chip can be easily separated. (*E*) Channel bonded at higher temperature (280 °C) with significant deformations.

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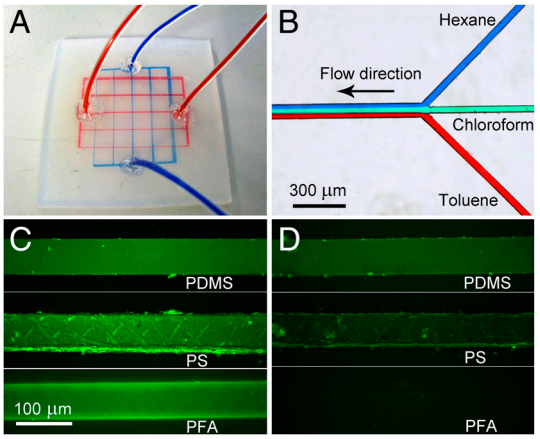


Fig. 3. Solvent-resistance (*A*, *B*) and antifouling property (*C*, *D*) of Teflon channels. (*A*) A PFA chip with two-layer microchannels separated by a thin FEP membrane that are filled with acetone (colored with a red dye) and DMSO (colored in a blue dye). (*B*) Laminar flow of dyed organic solvents in a whole-Teflon chip. (*C*) Fluorescence images of different kinds of microchannels filled with a 100 μg/mL GFP aqueous solution. (*D*) Fluorescence images of the channels in *C* after washing with buffer for 1 min.

after washing with buffer (Fig. 3*D*). This super cleanness of the Teflon microchannel is of great importance to quantitative analysis of microchips.

To broaden the applications of the Teflon chips, it is crucial for us to build integrated valves in them that control and regulate the flow of liquids in channels. Fortunately, both PFA and FEP are softer than common hard plastics; a thin membrane of FEP allows us to adopt the design of a widely used pneumatic valve (26) into our Teflon chips (Fig. 4). In the bonded chip, the control channel and fluidic channel are separated by a thin Teflon membrane (15-μm thick). When we apply pressure to the control channel the membrane deforms and blocks the fluidic channel. Compressed air at 0.1 MPa can completely close the valve (Fig. 4*B*). When the valve is open, the rate of flow is very high and increases linearly with pressure. In contrast, the flow through the closed valves was negligible (<100 pL/s at 10 kPa fluid driven pressure). By serially linking multiple valves together, we fabricated a nanoliter pump on the microchip (inset of Fig. 4*C*). The liquid is pumped forward when the valves close and open in designed sequence. The pumping rate is dependent on the operation frequency. For example, when the pump is operated

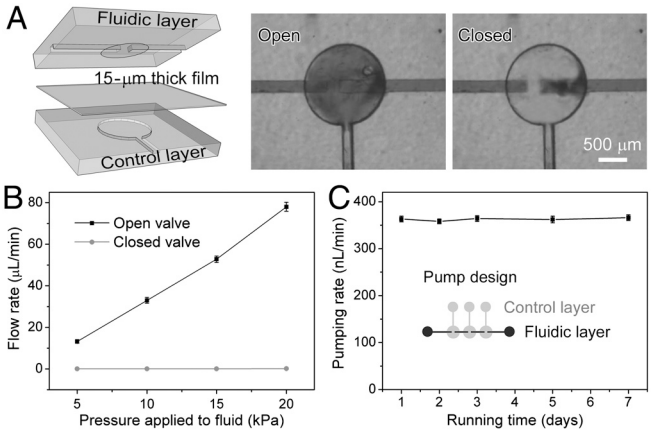


Fig. 4. (*A*) Exploded illustration and microphotographs of the whole-Teflon microvalve. We flowed 1 mM crystal violent in ethanol through the valve and used 0.1 MPa air pressure to control the valve. The FEP film seals against the gap in the fluidic channel when the valve is closed. (*B*) Flow rate versus driven pressure of the fluid through an open valve and a closed valve. (*C*) Long-term performance of a whole-Teflon pump. (*Inset*) Mask design of the pump.

at 1 Hz, the volume of fluid pumped per cycle is approximately 30 nL. Fig. 4*C* plots the pumped volumes during operation for 7 d, demonstrating excellent performance of the pump with high stability and long operation life. As the Teflon materials could maintain flexibility and mechanical strength over a temperature range from well below −100 °C to over 200 °C, (27) the whole-Teflon microvalves are expected to endure these extreme conditions.

We also studied the application of our Teflon microchips for long-term cell culture. In Fig. 5, we cultured a common cell line (HepG2 cells) in a PDMS channel (pretreated by oxidation in an oxygen plasma for 1 min) and a native PFA channel, respectively. Both channels are 100-μm wide, 100-μm deep, and 2-cm long. The cells are seeded at a density of 2×10^6 cells/mL. After the culture medium is infused into the channel, the ends of the tubing to the channels are sealed. The culture medium in the channel (and in the tubing) is enough for 120 h cell culture. We found that cells quickly settle down and spread out within 1 h, and can proliferate well for long-term culturing in both channels. After culturing for 120 h, there is no significant difference between the cells in the middle part and at the ends of the channels. Our results show that the cells proliferate at similar rates in the PFA channel and in the PDMS channel (Fig. S2). This finding suggests that the gas permeability of the Teflon materials [$\sim 7 \times 10^{-4}$ μg/(cm^2/cm-min) for O_2 and $\sim 1 \times 10^{-6}$ μg(cm^2/cm-min) for CO_2] (available at http://www2.dupont.com/Teflon_Industrial/en_US/index.html; http://www.arkema-inc.com/literature/pdf/775.pdf) is sufficient for long-term cell culturing inside these Teflon microchannels (28). Because of the nonsticking property of Teflon, it is easy to flush out lysed and dead cells completely from the channel (Fig. 5*E*). Unlike microchannels fabricated using PDMS or PS, the Teflon channels is not contaminated and can be reused. Fig. 6 shows the culturing of HeLa cells in a 100-μm wide, 100-μm deep, and 2-cm long PFA channel for 5 d. Our results suggest that the Teflon chip is well suited for long-term cell culturing.

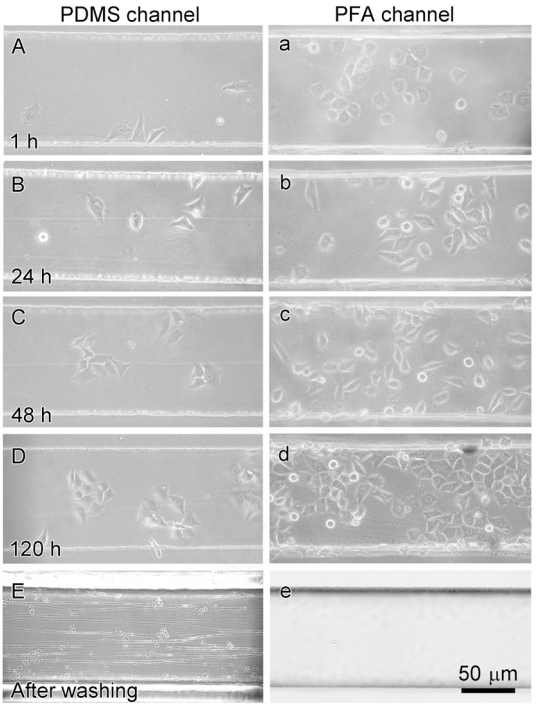


Fig. 5. Comparison of HepG2 cells that are cultured in PDMS channel (*Left*) and PFA channel (*Right*). Both the PDMS and the PFA channels are 100-μm wide, 100-μm deep, and 2.0-cm long. The bottom row of microphotographs shows the microchannels after the cells being cultured for 120 h in the channels are lysed with 0.1 M NaOH and flushed with PBS solution for 5 min, respectively.

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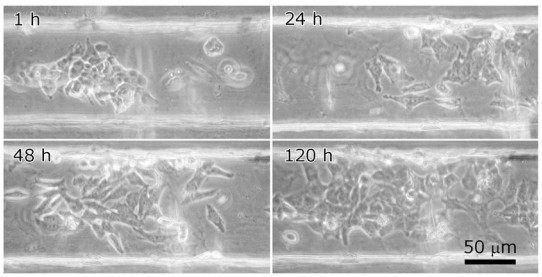


Fig. 6. Five-day culture of HeLa cells in a 100-μm wide, 100-μm deep, and 2.0-cm long Teflon PFA channel.

Conclusions

In summary, we demonstrated a convenient method to fabricate whole-Teflon microfluidic chips with integrated microvalves and pumps. These chips are optically transparent and are resistant to solvents and surface fouling. Similar to PDMS chips, biological cells can proliferate well in the Teflon microchannels. Because small molecules cannot be absorbed into Teflon and no molecules leach from Teflon (unlike PDMS), Teflon chips may give a better platform than PDMS chips for cell culturing and cell-based drug screening researches. In addition, both Teflon plates and thin films are commercially available with prices ($\sim$ 20–40/kg) comparable to PDMS (typically our Teflon chip is $\sim$2 mm in thickness). By combining the ease of fabrication and the special properties of Teflon materials, we expect that they can be the materials for the next generation of microfluidic chips and will greatly expand the applications of microfluidics to a wider range of substances.

Materials and Methods

Materials and Equipment. We obtained PDMS prepolymer and GFP from GE Silicones RTV615 (GE) and Invitrogen, respectively. We purchased PFA (approximately 1-mm thick plate) and FEP (15-μm thick membrane) from Yuyisong, Inc. All other chemicals were from Sigma-Aldrich that were of analytical grade and used without further purification. All photoresists were purchased from Microchem.

We performed thermo molding of the Teflon substrates on a hot compressor (TM-101F, Taiming, Inc.). Baking of the PDMS masters and bonding of the Teflon chips were processed with an oven (KSW5-12-A, Zhonghuan Experiment Electric Stove). SEM images were taken using a JEOL electronic microscope (JEOL Ltd.). Images of solvent compatibility experiments and valve operation process were captured using a microscope (AZ100, Nikon) equipped with a cooled charge-coupled device camera (DS-Fi1, Nikon). Images of cells cultured in microchannels were taken using an inverse microscope (Nikon Eclipse TE2000-U) with a computerized cooled charge-coupled device camera (Diagnostic Instruments). For laminar flow generation in the microchannel, fluids were driven by syringe pumps (Harvard Pump 11 Plus Dual Syringe, Harvard Apparatus) equipped with latex-free syringes (NORM-JECT).

Preparation of PDMS Masters. Photoresist (SU-8 and AZ series) microstructures were fabricated with standard photolithography. Positive photoresist structures were heated on a 150 °C hot plate for 5 min to reflow the photoresist and round the photoresist profile (29).

We spincoated an approximately 500-μm thick layer of PDMS prepolymer (the weight ratio of component A and B is 5 : 1) onto the photoresist patterns. After degassing in a vacuum chamber, we heated the PDMS in a 70 °C oven for approximately 30 min. We peeled off the cured PDMS membrane and placed it onto a piece of glass with its pattern surface exposed. We heated the PDMS-on-glass structure in a 250 °C oven for 1 h. The PDMS membrane are fully cured and tightly sealed to the glass slide. This PDMS structure on the glass slide was used as the master to mold Teflon substrates.

Fabrication of Teflon Microchips. We sandwiched the PFA substrate between the PDMS master and another flat glass slide (coated with a thin PDMS layer). We placed this sandwich assembly on a hot compressor (TM-101F, Taiming, Inc.) and embossed at 275 °C for 2 min (a wide range of pressure in the tens of kPa could be used), then removed the assembly from the working stage of the equipment and placed it between two CPU coolers to rapidly cool down to room temperature. In this way, the molding process takes approximately 5 min/piece. Because of slightly lower glass transition point of FEP, its molding temperature should be adjusted to 265 °C.

Before bonding, holes were drilled through the Teflon plates for reservoirs and tubing connections; the surfaces to be bonded were cleaned by acetone and blown dry. We assembled the Teflon substrates to be bonded between two pieces of 5-mm thick glass plates that were coated with an approximately 1-μm thick PDMS layer. We fixed the assembly with four screw clamps, slightly tightened (figure-tight) the screws and placed it in a 260 °C oven for 1 h. After cooling down to room temperature, the Teflon substrates were bonded and the bonded chip was connected through PFA tubing.

Organic solvents can fill the Teflon microchannels without trapping bubbles. To avoid trapping air bubbles while filling in aqueous solutions, we filled the channel with 70% weight alcohol followed by 30% weight alcohol and then the wanted aqueous solution. In this way, all the space in the channel could be completely filled with the aqueous solution. A solenoid valve array (N37, Yong Jing) controlled by a LabVEIW program was used to regulate the air pressure in the controlling channels for operating the pneumatic valves and pumps.

Culturing Cells on Teflon Chips. Liver cancer cell line HepG2 and cervical cancer cell line HeLa were purchased from American Type Culture Collection, handled under a sterile tissue culture hood and maintained in Dulbecco's modified Eagle minimum essential medium and Eagle minimum essential medium, respectively. Both media were supplemented with 10% weight FBS, 100 u/mL penicillin and 100 μg/mL streptomycin (all were purchased from Gibco, Invitrogen). The cells were cultured in a cell incubator at 37 °C with 5% CO_2.

Before experiments, we sterilized the Teflon devices using 70% weight ethanol aqueous solution for 0.5 h, and sequentially replaced the ethanol solution by 50% weight ethanol, 30% weight ethanol, and PBS aqueous solution to prevent bubble trapping in the channels. We sterilized the PDMS devices under UV light for 0.5 h. The cells were passaged with standard 0.25% weight trypsin solution (Gibco, Invitrogen), which was neutralized by medium. The cell suspension solution was centrifuged at 1,250 rpm for 3 min and the cells were resuspended in cell culture medium. After counting the cell number on a hemocytometer, we diluted the cells to a density of 2×10^6 cells/mL and injected them to the microchannels with a syringe pump for further culture. After 5 d culture, a 0.1 M NaOH solution was infused into both PDMS and Teflon channels and left in the channels for 5 min to lyse the cells. Finally, we washed the microchannels with PBS solution.

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